

## Tekstmodel

|    | model                                                                                | initial values                                |
|----|--------------------------------------------------------------------------------------|-----------------------------------------------|
| 1  | $I = U_b / (R_3 + R_{1,2}/2)$                                                        | $t = 0 \text{ s}$                             |
| 2  | $U_{1,2} = I \cdot R_{1,2}/2$                                                        | $dt = 1 \times 10^{-4} \text{ s}$             |
| 3  | $U_3 = I \cdot R_3$                                                                  | $U_b = 18 \text{ V}$                          |
| 4  | $P_{1,2} = I^2 \cdot R_{1,2}/4$                                                      | $R_0 = 5.0 \Omega$                            |
| 5  | $P_3 = I^2 \cdot R_3$                                                                | $T_0 = 293 \text{ K}$                         |
| 6  | $T_{1,2} = T_{1,2} + (P_{1,2} - \varepsilon \sigma A \cdot T_{1,2}^4) \cdot dt / mc$ | $\alpha = 5.2 \times 10^{-3} \text{ K}^{-1}$  |
| 7  | $T_3 = T_3 + (P_3 - \varepsilon \sigma A \cdot T_3^4) \cdot dt / mc$                 | $\varepsilon \sigma A = 7.14 \times 10^{-14}$ |
| 8  | $R_{1,2} = R_0 \cdot (1 + \alpha \cdot (T_{1,2} - T_0))$                             | $mc = 3 \times 10^{-5} \text{ J K}^{-1}$      |
| 9  | $R_3 = R_0 \cdot (1 + \alpha \cdot (T_3 - T_0))$                                     | $I = 0 \text{ A}$                             |
| 10 | $t = t + dt$                                                                         | $R_{1,2} = 5.0 \Omega$                        |
| 11 | IF $t \geq 0.3$ THEN STOP ENDIF                                                      | $R_3 = 5.0 \Omega$                            |
|    |                                                                                      | $U_{1,2} = 0 \text{ V}$                       |
|    |                                                                                      | $U_3 = 0 \text{ V}$                           |
|    |                                                                                      | $P_{1,2} = 0 \text{ W}$                       |
|    |                                                                                      | $P_3 = 0 \text{ W}$                           |
|    |                                                                                      | $T_{1,2} = 3 \times 10^2 \text{ K}$           |
|    |                                                                                      | $T_3 = 3 \times 10^2 \text{ K}$               |

## Grafisch model

